UNIVERSITY OF LONDON

BSc EXAMINATION 2024

For Internal Students of
Royal Holloway

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CS3003: IT Project Management
CS3003R: IT Project Management – for FIRSTSIT/RESIT CANDIDATES

Time Allowed: **TWO** hours

Please answer **ALL** questions

Calculators are not permitted

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1.

(a) Indicate whether each of the following statements is **true** or **false**. Briefly **justify** your answers.

(i) Change requests are a sign of poor project planning and should generally be avoided.

(ii) Parametric estimation methods depend on the existence of a historic model.

(iii) The scope of a project should always be clear and comprehensive before the initiation phase commences.

(iv) In configuration management, software documentation is a valid configuration item.

(v) It is an appropriate practice for different projects to use their own, project-specific, source code version management systems.

(vi) The “Issue Log” in an alternative name for a project’s “Risk Register”.

[12 marks]

(b) In the context of quality management, explain the difference between product quality and project process quality. [4 marks]

(c) You are the Project Manager in a software development project that will deliver a new e-commerce system. The project has a very clear and detailed scope, which was rigorously defined and agreed in the planning phase. During the project’s execution phase, a senior engineer calls you and suggests a more efficient design to handle the user authentication part of the system. The engineer’s design could benefit the project. Which of the following is the most appropriate action to take? **Justify** your answer.

- Do nothing, design changes cannot be accommodated at the execution phase given that the project scope was rigorously defined and agreed in the planning phase
- Discuss the design change with the project sponsor
- Discuss the design change with the entire team
- Ask the engineer to submit a relevant change request
- Evaluate the impact of the proposed design change on the project constraints

[10 marks]

(d) Draw a diagram depicting the four generic activities of the risk management process. Outline the tasks performed in each activity and the respective outcomes.

[8 marks]

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2.

(a) You are the Project Manager in a project that is expected to run for two months and consists of two major activities, namely Activity A and Activity B. The actual cost of Activity A is £10,000 and the actual cost of Activity B is £20,000. The planned value for these activities is £8,000 and £18,000 respectively. Activity B is 100% complete. However, Activity A is only 75% complete.

Using **exclusively earned-value management**, indicate which of the following statements is correct about the project. **Justify** your answer.

- The project is behind schedule and under budget
- The project is behind schedule and over budget
- The project is ahead of schedule and over budget
- The project is ahead of schedule and under budget

[10 marks]

(b) The activities of a project and respective durations and dependencies are shown in the following table.

Activity	Duration (weeks)	Predecessors
A	3	-
B	10	A
C	8	A
D	11	A
E	1	B,C
F	2	C
G	2	E,F
H	1	G,D

(i) Draw a network diagram for this project including all floats. [9 marks]

(ii) Mark the project's critical path on the diagram. [2 marks]

(c) One of the Project Manager's responsibilities is to manage project communications.

(i) Provide the definition of Project Communications Management.

[4 marks]

(ii) Refer to any four elements that a Project Communications Plan should include. [4 marks]

(iii) Describe any two actions a Project Manager can take in order to develop an appropriate Project Communications Plan. [4 marks]

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3.

(a) A company is about to buy a software package, Package A, that consists of two components, namely A1 and A2. These components are licensed independently according to the following scheme:

- A1 is licensed on a per user basis, costing £200 per user.
- A2 is licensed based on the number of CPUs (processors) that are installed on the server that Package A uses. The license cost of A2 is £10,000 per installed CPU. It is estimated that in order to adequately perform in the production environment, Package A requires 4 CPUs for up to 300 users, 8 CPUs for 301 to 600 users and 16 CPUs for 601 to 1000 users. There are fewer than 1000 users in the company.

Moreover, starting from the second year, the company is obliged to pay the vendor of Package A an annual maintenance fee of 20% over the license fee. Finally, each CPU of the production environment is acquired at a cost of £5,000 and has an annual maintenance fee of 10% over the acquisition cost. The CPU maintenance fee also starts from the second year.

If variable N denotes the number of users and variable M the number of CPUs, based on the previous facts, provide a single formula to calculate the 5-year Total Cost of Ownership (TCO) of Package A. The TCO formula should be of the form TCO= aN + bM + c, where a, b and c are numbers.

Notes: Assume that there are no other costs for Package A. The time-value of money should not be taken into account in the TCO calculation.

[13 marks]

(b) Discuss the 5 different stages of the Tuckman model of team development.

[10 marks]

(c) You are the Project Manager in a software development project. One of the software developers informs you that she added an additional feature in the end-product, which was not included in the agreed specification. She believes the customer will really appreciate the feature added. Which of the following is the most appropriate action to take? **Justify** your answer.

- Ask the developer to remove the additional feature from the existing software build and, going forward, ensure that the project team develops the software based on the agreed specification and the project's established processes
- Thank the developer for the extra effort, accept the additional feature and ask the test coordinator to include it in the UAT test cases
- Accept the feature and update the project's cost baseline given that the project may run over budget due to the feature addition

[10 marks]

END

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